



Relative left frontal activity in reappraisal and suppression of negative emotion: Evidence from frontal alpha asymmetry (FAA)



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ABSTRACT

Previous studies have shown that reappraisal (changing the way that one thinks about emotional events) is an effective strategy for regulating emotion, compared with suppression (reducing emotion-expressive behavior). In the present study, we investigated relative left frontal activity when participants were instructed to use reappraisal and suppression of negative emotion, by measuring frontal alpha asymmetry (FAA). Two electroencephalography (EEG) experiments were conducted; FAA was analyzed while 102 healthy participants (59 men, 43 women) watched negative images after being instructed to perform reappraisal (Experiment 1) and suppression (Experiment 2). Habitual use of reappraisal and suppression was also assessed using the emotion regulation questionnaire (ERQ). The results of Experiment 1 showed that relative left frontal activity was greater when instructed to use reappraisal of negative images than when normally viewing negative images. In contrast, we observed no difference between conditions of instructed suppression and normal viewing in Experiment 2. In addition, in male participants, habitual use of reappraisal was positively correlated with increased relative left frontal activity for instructed reappraisal, while habitual use of suppression did not show a significant correlation with changes in relative left frontal activity for instructed suppression. These results suggest that emotional responses to negative images might be decreased for instructed reappraisal, but not suppression. These findings support previous reports that reappraisal is an effective emotion regulation strategy, compared with suppression.

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1. Introduction

Emotions are regulated in various ways in everyday life. The most commonly investigated strategies for regulating emotion are cognitive reappraisal (hereafter, reappraisal) and expressive suppression (hereafter, suppression). Reappraisal refers to the act of changing the way one thinks about emotional events, while suppression refers to reducing emotion-expressive behavior (reviewed in John & Gross, 2004). A number of studies have reported that these two strategies have different effects on both psychological and physiological responses (for example, Gross, 1998; Butler et al., 2003; Goldin et al., 2008; McRae et al., 2012). For example, subjective experience of negative emotion was reported to be decreased by reappraisal, but unchanged by suppression (Gross, 1998). In addition, habitual use of reappraisal is related to positive affectivity and better psychological well-being, while habitual use

of suppression is related to negative affectivity and poorer psychological well-being (Gross and John, 2003; McRae et al., 2012). A functional magnetic resonance imaging (fMRI) study reported that responses in emotion-generative regions such as the amygdala and insular to negative stimuli were decreased by reappraisal, and increased by suppression (Goldin et al., 2008). Taken together, these results suggest that reappraisal is a more effective and healthier strategy for emotion regulation than suppression, as suggested by John & Gross (2004) and Augustine and Hemenover (2009).

Several previous electroencephalography (EEG) studies reported that automatic emotion regulation is related to greater relative left frontal activity, as measured by frontal alpha asymmetry (FAA; Jackson et al., 2003; Goodman et al., 2013). In a study by Jackson et al. (2003), participants viewed negative stimuli without performing any specific emotion regulation strategy (e.g., reappraisal or suppression). The results revealed that individuals with greater baseline relative left frontal activity (measured by the FAA) showed greater attenuation of eyeblink startle responses after viewing negative stimuli. Goodman et al. (2013) found a similar relationship between greater transient relative left frontal activity (measured by FAA) and attenuation of eyeblink startle responses under stress. Given that attenuation of eyeblink startle responses after viewing emotional stimuli reflects a decreased

Abbreviations: fMRI, functional magnetic resonance imaging; EEG, electroencephalography; FAA, frontal alpha asymmetry; ERP, event-related potential; ERQ, emotion regulation questionnaire; ANOVA, analysis of variance; SD, standard deviation; SE, standard error.

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emotional response (Lang et al., 1990; Van Reekum et al., 2011), the results of Jackson et al. (2003) and Goodman et al. (2013) suggest that increased relative left frontal activity (measured by both state and trait FAA) is related with decreased emotional response to negative stimuli through automatic emotion regulation. This explanation is in line with the widely accepted relationship between FAA and the approach-avoidance motivational system; greater relative left frontal activity involves mainly positive emotion related with approach motivation (e.g., joy), whereas right-sided frontal activity involves negative emotion related with avoidance motivation (e.g., fear and disgust; reviewed in Davidson, 1998; Davidson, 2004; Harmon-Jones et al., 2010).

Taken together, these findings suggest that greater relative left frontal activity may be related to decreased emotional response to negative stimuli not only through automatic emotion regulation (Jackson et al., 2003; Goodman et al., 2013), but also through emotion regulation strategies, such as reappraisal and suppression. Several recent EEG studies have investigated left frontal activity during reappraisal and suppression (Parvaz et al., 2012; Tolegenova et al., 2014; Wang et al., 2015; Papousek et al., 2016); however, the results are mixed. Parvaz et al. (2012) indicated that reappraisal of negative images and emotionally neutral images increased left frontal, but not right frontal, activity. Papousek et al. (2016) found a positive relationship between relative left frontal activity and reappraisal ability during reappraisal of anger-evoking events, as well as greater relative left frontal activity during a reappraisal task compared with the resting condition at ventrolateral prefrontal electrodes. Tolegenova et al. (2014) reported a significant relationship between emotional intelligence and EEG responses during reappraisal and suppression of fear-evoking film clips in frontal theta and gamma power, but not in frontal alpha power. These results suggest the importance of further examination of relative left frontal activity during reappraisal and suppression.

The present study aimed to investigate relative left frontal activity when participants were instructed to perform reappraisal and suppression of negative emotion, by measuring FAA. We conducted two experiments based on a protocol used in a recent event-related potential (ERP) study by Moser et al. (2014), which found that instructing participants to perform reappraisal decreased emotional arousal responses to negative images. In Experiment 1, we measured FAA while healthy young participants watched negative images after being instructed to perform reappraisal and observation. In Experiment 2, we measured the FAA while the same participants viewed negative images after being instructed to perform suppression and observation. We hypothesized that relative left frontal activity would be greater while performing reappraisal, compared with observation (Experiment 1) and that relative left frontal activity would not differ between suppression and observation (Experiment 2). This hypothesis is in accord with previous findings suggesting that reappraisal is a more effective emotion regulation strategy for decreasing emotional responses to negative stimuli compared with suppression (Gross, 1998; Butler et al., 2003; Goldin et al., 2008; McRae et al., 2012; McRae et al., 2012).

Additionally, the present study focused on the relationship between relative left frontal activity and habitual use of emotion regulation (measured with the emotion regulation questionnaire, ERQ, Gross and John, 2003). Several previous studies reported that habitual use of reappraisal affects emotional responses during reappraisal (Drabant et al., 2009; Grecucci et al., 2013; Moser et al., 2014). For instance, an fMRI study by Drabant et al. (2009) showed that the habitual use of reappraisal was positively correlated with decreased activation in emotion-generative regions such as the amygdala, suggesting a relationship between the habitual use of reappraisal and effective emotion regulation. Thus, in Experiment 1 of the present study, we hypothesized that individuals exhibiting greater habitual use of reappraisal would show a greater increase in relative left frontal activity when instructed to perform reappraisal than those with low habitual use of reappraisal. In contrast, there have been few studies of the relationship between the habitual use of suppression and emotional responses to

negative stimuli. Meyer et al. (2012) showed that a decreased subjective experience of emotional arousal to negative stimuli was related to habitual use of reappraisal, but not habitual use of suppression. This suggests that habitual use of suppression might be unrelated to emotion regulation responses. Based on this finding, in Experiment 2 of the present study, we hypothesized that there would be no relationship between the habitual use of suppression and relative left frontal activity changes when participants were instructed to perform suppression.

2. Methods

2.1. Participants

A total of 111 Japanese undergraduate and graduate students (62 men, 49 women; age range: 19–25 years; all right-handed) participated in Experiment 1 and Experiment 2.¹ Participants had normal or corrected-to-normal vision and were not taking any prescription medications. EEG quality was poor for nine participants, leaving 102 individuals (59 men, 43 women) for the final analysis. Participants provided written informed consent before participation, after receiving a brief explanation of the study.² The study protocols were approved by the ethics committee of Kyushu University, Japan.

2.2. Habitual use of emotion regulation

Participants filled out the Japanese version (Yoshizu et al., 2013) of the ERQ (Gross and John, 2003). The ERQ contains 6 items for the reappraisal scale (e.g., “I control my emotions by changing the way I think about the situation I’m in”) and 4 items for the suppression scale (e.g., “I control my emotions by not expressing them”). Participants rated on a Likert-scale from 1 (“does not describe me well”) to 7 (“describes me very well”). Internal consistency (Cronbach’s alpha) was 0.686 for ERQ-reappraisal and 0.753 for ERQ-suppression.

2.3. Stimuli

In Experiment 1, 60 images (30 neutral and 30 negative) were selected from the International Affective Picture System (IAPS; Lang et al., 2008). The neutral images included scenes such as a man with an emotionally neutral expression, or basic objects such as a book or a clock. Because the neutral images were included as a filler to prevent possible habituation, as in Moser et al. (2014), the responses to neutral images were not analyzed. The negative images included scenes such as those depicting violence between people, an injured animal, or an airplane accident. Normative ratings for the IAPS (Lang et al., 2008) indicated that the negative images we chose were more unpleasant and more arousing than the neutral ones (valence rating: 4.97 for neutral images, 2.33 for negative images; arousal rating: 2.98 for neutral images, 6.27 for negative images). The specific IAPS picture identification numbers are presented in the Appendix A. The images used in Experiment 1 were the same as the images used in Moser et al. (2014).

In Experiment 2, a different set of 60 images (30 neutral and 30 negative) were selected from the IAPS (Lang et al., 2008), to prevent possible habituation to the images presented in Experiment 1. Normative ratings for the IAPS (Lang et al., 2008) were as follows: valence rating: 5.05 for neutral images, 2.38 for negative images; arousal rating: 3.24 for neutral images, 6.20 for negative images. An independent *t*-test revealed no significant differences in the normative valence or arousal ratings between the images used in Experiment 1 and Experiment 2.

¹ Three female participants participated only in Experiment 1. They were excluded from the final analysis, enabling comparison between results of Experiment 1 and Experiment 2.

² To assess personality characteristics of participants, we also asked participants to complete the short Japanese version of the Big Five scale (Namikawa et al., 2012), which was developed based on an adjective checklist scale (Piedmont et al., 1991). For brevity, the results of the Big Five scale are not reported here.

(valence rating for neutral images: $t = 0.92$, $p = 0.364$; valence rating for negative images: $t = 1.36$, $p = 0.179$; arousal rating for neutral images: $t = 0.38$, $p = 0.704$; arousal rating for negative images: $t = -0.47$, $p = 0.643$; all $df = 58$).

2.4. Procedure

The experimental procedure was based on that of Moser et al. (2014). In Experiment 1, participants sat on chairs approximately 80 cm from a 20-inch LCD monitor. After EEG sensors were attached, participants were given detailed task instructions, as follows: 1) negative or neutral images will be presented on the monitor; 2) before the images appear, one of two instructions (“Observe” or “Reappraise”) will be presented; 3) after being instructed to “observe”, you should respond naturally to the images, and, after being instructed to reappraise, you should try to think of the image in a positive way so that the intensity of your negative emotions decreases; 4) you should not generate thoughts that are completely unrelated to the presented images (for example, the beginning of a holiday) while watching the images. We provided participants with some example sentences for reappraisal (for example, “a bloody crime scene could be seen as the place where a murder investigation was finally solved” [Hajcak and Nieuwenhuis, 2006]).

The experimental block consisted of three conditions (Neutral, Negative, and Reappraise Negative) that differed in terms of the type of image presented (neutral or negative) and in the instructions (“Observe” or “Reappraise”). In each trial, the instruction phrase was first presented for 2 s. “Observe” was presented for the Neutral and Negative conditions and “Reappraise” for the Reappraise Negative condition. The instruction phase was followed by a blank screen for 0.5 s and then by a central fixation cross for 0.5 s. The fixation cross was followed by a 6-s presentation of an image stimulus. Each condition was presented for 30 trials (inter-trial interval: 2.5 s). Each neutral image was shown once and each negative image was shown twice (once in the Negative condition and once in the Reappraise Negative condition). The order of trials was random. A practice block (five trials of each condition) using a different set of 15 images was administered before the experimental block.

The procedure for Experiment 2 was the same as that of Experiment 1, except that the Reappraise Negative condition was replaced with a Suppress Negative condition. Participants were given detailed task instructions as follows: 1) negative or neutral images will be presented on the monitor; 2) before the images appear, one of two instructions (“Observe” or “Suppress”) will be presented; 3) after being instructed to “observe”, you should respond naturally to the images, while after being instructed to “suppress”, you should try to suppress your negative emotions and hide emotional reactions to the images so that we will not be able to detect what kind of image you are viewing (your facial expressions will be monitored); 4) you should not generate thoughts that are completely unrelated to the presented images (for example, the beginning of a holiday) while watching the images. The instructions in the Negative Suppress condition were based on those used by Murata et al. (2013).

2.5. EEG measurement and analysis

EEG signals were recorded using a 64-channel Geodesic Sensor Net and Net Station (Electrical Geodesics, Inc., Eugene, OR, USA) based on the 10/20 system and were amplified by a high-input impedance (200 M Ω) amplifier (Net Amps 200 Amplifier, Electrical Geodesics, Inc.). During recording, EEG signals were recorded at electrode site Cz as a reference with a sampling frequency of 500 Hz. Electrode impedances were maintained below 50 k Ω .³

³ We note that impedances in the 50 k Ω range may not compromise EEG quality when the input impedance of the amplifier is high (>100 M Ω) (Allen et al., 2004).

After recording, EEG data were re-referenced to the average of the left and right mastoids and band-pass filtered with cutoffs of 0.1 Hz and 45 Hz, using EMSE software (Source Signal Imaging Inc., San Diego, CA, USA). The spectral power during image presentation (6 s) was then obtained through a Fast Fourier transform (FFT) (epoch length 2 s, 50% overlap, Hanning window). Epochs including artifacts caused by eye blinks, muscle twitches, or body movements were rejected if they were greater than ± 100 μ V. The mean numbers and standard deviations of epochs used for analysis were as follows; 119.9 ± 24.4 for the Negative condition in Experiment 1; 121.5 ± 24.2 for the Reappraise Negative condition in Experiment 1; 118.2 ± 24.8 for the Negative condition in Experiment 2; 121.7 ± 25.3 for the Suppress Negative condition in Experiment 2.

Power within the alpha band (8–13 Hz) at each site was then natural-log transformed. Given that FAA is usually assessed by subtracting alpha power in the left frontal area from that in the right frontal area (for example, Sutton and Davidson, 1997; Harmon-Jones and Sigelman, 2001; Coan and Allen, 2003a, 2003b), the present study calculated FAA from the three pairs of frontal electrodes using the following equation: fronto-polar sites: $\ln(\text{Fp2})$ minus $\ln(\text{Fp1})$; mid-frontal sites: $\ln(\text{F4})$ minus $\ln(\text{F3})$; lateral-frontal sites: $\ln(\text{F8})$ minus $\ln(\text{F7})$. Because alpha power is inversely related to brain activity (Cook et al., 1998), greater FAA obtained by this equation represents greater relative left frontal activity.

2.6. Statistical analysis

For ERQ scores, correlation analysis was conducted between ERQ-reappraisal and ERQ-suppression (Pearson correlation coefficient). Gender differences in ERQ scores were also examined (independent t -tests). For the FAA, a repeated-measures ANOVA was conducted on FAA with *Condition* (Experiment 1: Negative vs. Reappraise Negative; Experiment 2: Negative vs. Suppress Negative) and *Site* (fronto-polar, vs. mid-frontal, vs. lateral-frontal region) as within-subject factors and *Gender* as a between-subject factor. The Greenhouse-Geisser correction was applied where sphericity was violated. When a main effect or interaction was significant, post-hoc comparisons were performed with Bonferroni correction. In addition, we examined correlations between ERQ scores and FAA differences between conditions (Experiment 1: Reappraise Negative minus Negative; Experiment 2: Suppress Negative minus Negative) (Pearson correlation coefficient). Statistical significance was set at ($p < 0.05$). Analyses were conducted using SPSS software (Version 17, SPSS Inc., Chicago, IL, USA).

3. Results

3.1. ERQ scores

Table 1 shows the mean scores and range of ERQ scores. Mean ERQ scores in the present study were similar to those of Japanese University students reported in a previous study (ERQ-reappraisal: Mean = 4.63, SD = 1.0; ERQ-suppression: Mean = 3.96, SD = 1.16; Matsumoto et al., 2008). No significant correlation was found between ERQ-

Table 1
Mean and range of emotion regulation questionnaire (ERQ) scores.

	Mean (SD)	Range
ERQ-reappraisal		
All participants	4.73 (0.81)	2.00–6.50
Male participants	4.81 (0.82)	2.50–6.50
Female participants	4.64 (0.82)	2.00–6.00
ERQ-suppression		
All participants	3.65 (1.11)	1.25–6.75
Male participants	4.00 (1.10)	1.75–6.75
Female participants	3.19 (0.96)	1.25–5.25

Note. SD: standard deviation; all participants: N = 102; male participants: N = 59; female participants: N = 43; ERQ: emotion regulation questionnaire.

Table 2

Analysis of variance (ANOVA) effects of frontal alpha asymmetry (FAA). Significant *p*-values are in bold.

	<i>df</i> (h,e)	Experiment 1		Experiment 2	
		<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>
Condition	1, 100	7.28	0.008	0.22	0.638
Site	2, 200	7.16	0.002	12.98	0.000
Gender	1, 100	0.08	0.783	2.11	0.150
Condition × site	2, 200	2.80	0.074	0.76	0.449
Condition × gender	1, 100	0.41	0.524	0.47	0.493
Site × gender	2, 200	2.94	0.065	6.74	0.002
Condition × site × gender	2, 200	1.04	0.345	0.19	0.785

Note. *df*(h,e): degrees of freedom. The Greenhouse-Geisser correction was applied where sphericity was violated; the uncorrected degrees of freedom are reported in this table.

reappraisal and ERQ-suppression scores ($r = 0.018$, $p = 0.857$, Pearson correlation coefficient).

The results revealed no significant gender difference in ERQ-reappraisal score ($t = 0.99$, $df = 100$, $p = 0.326$, independent *t*-test). Meanwhile, ERQ-suppression showed a significant gender difference ($t = 3.86$, $df = 100$, $p < 0.001$, independent *t*-test), indicating that male participants showed higher ERQ-suppression score than female participants.

3.2. FAA

Table 2 shows a summary of the statistical results of the FAA. Table 3 shows the mean FAA in each pair of sites.

In Experiment 1, the repeated-measures ANOVA revealed a main effect of Condition (Table 2), indicating that the FAA was significantly greater in the Reappraise Negative condition than in the Negative condition as shown in Fig. 1A. The main effect of Site was also significant (Table 2); post-hoc comparisons with a Bonferroni correction showed that the FAA was significantly greater in mid-frontal compared with fronto-polar ($t = 3.18$, $df = 203$, corrected $p = 0.006$) and lateral-frontal sites ($t = 3.31$, $df = 203$, corrected $p = 0.004$).

In Experiment 2, a repeated-measures ANOVA revealed no main effect of Condition or a Condition × Site interaction (Table 2), indicating that the FAA was not significantly different between the Suppress Negative condition and the Negative condition (Fig. 1B). The main effect of Site and the Site × Gender interaction were significant (Table 2); post-hoc comparisons with a Bonferroni correction showed that the FAA was significantly greater at mid-frontal compared with fronto-polar ($t = 4.58$, $df = 85$, corrected $p < 0.001$) and lateral-frontal sites ($t = 5.03$, $df = 85$, corrected $p < 0.001$) in female participants, but not in male participants (all corrected $p > 0.05$).

3.3. Correlations between FAA and ERQ scores

In Experiment 1, ERQ-reappraisal was not significantly correlated with FAA differences (Reappraisal Negative minus Negative) averaged across three pairs of sites ($r = 0.187$, $p = 0.060$, Fig. 2A). The correlation between ERQ-reappraisal and FAA difference was also tested in male and female participants separately. The results revealed that ERQ-reappraisal was significantly positively correlated with FAA difference averaged across three pairs of sites in male participants ($r = 0.352$, $p = 0.006$, Fig. 2B),⁴ but not in female participants ($r = -0.039$, $p = 0.802$, Fig. 2C).

⁴ The correlations between ERQ-reappraisal and FAA difference (Reappraisal Negative minus Negative) at each pair of sites were analyzed. To adjust for multiple comparisons, the modified Bonferroni correction was performed online (<http://www.quantitativeskills.com/sisa/calculations/bonfer.htm>). The results revealed that, in male participants, ERQ-reappraisal was significantly positively correlated with FAA difference at mid-frontal and lateral-frontal sites, but not at fronto-polar sites (adjusted critical *p*-value = 0.0229; fronto-polar: $r = 0.265$, $p = 0.0428$; mid-frontal: $r = 0.303$, $p = 0.0195$; lateral-frontal: $r = 0.298$, $p = 0.0219$).

In Experiment 2, ERQ-suppression was not significantly correlated with FAA difference (Suppress Negative minus Negative) averaged across three pairs of sites ($r = -0.117$, $p = 0.243$, Fig. 2E). The correlation between ERQ-suppression and FAA difference was also tested in male and female participants separately. The results revealed that ERQ-suppression was not significantly correlated with FAA difference in male participants ($r = -0.186$, $p = 0.159$, Fig. 2E) or female participants ($r = 0.025$, $p = 0.876$, Fig. 2F).

4. Discussion

The present study investigated relative left frontal activity when participants were instructed to perform reappraisal and suppression of negative emotion, by measuring FAA. In addition, we examined whether relative left frontal activity when instructed to perform reappraisal and suppression were related to individuals' habitual use of reappraisal and suppression.

4.1. Effect of reappraisal on relative left frontal activity

The results of Experiment 1 indicated that relative left frontal activity was greater when instructed to perform reappraisal, compared with normal viewing. Previous studies have reported that greater relative left frontal activity is related to decreased emotional responses to negative stimuli through automatic emotion regulation (Jackson et al., 2003; Goodman et al., 2013) and to positive emotion based on approach motivation (reviewed in Davidson, 1998; Davidson, 2004; Harmon-Jones et al., 2010). Thus, the present findings indicate that being instructed to perform reappraisal may have decreased emotional responses to negative images. This interpretation is in line with previous studies of brain activity during reappraisal. Reappraisal was found to decrease responses in emotion generative regions such as the amygdala and insular (Goldin et al., 2008), and to decrease arousal level measured by ERP responses (Moser et al., 2014) when viewing negative stimuli. Thus, the present findings support previous reports that reappraisal is an effective strategy for emotion regulation (John & Gross, 2004; Augustine and Hemenover, 2009). However, the findings of the present study and previous studies (Goldin et al., 2008; Moser et al., 2014) are from laboratory tasks. Thus, it is unclear whether this effect of reappraisal on emotional response is also shown in response to negative events that occur in everyday life. Future research is needed to examine whether increased relative left frontal activity is exhibited during reappraisal of real-life experiences.

The present finding may also be helpful for understanding the mixed results of previous EEG studies that examined relative left frontal activity during reappraisal (Parvaz et al., 2012; Tolegenova et al., 2014; Wang et al., 2015; Papousek et al., 2016). Previous studies reported an effect of reappraisal of negative stimuli on increased left frontal activity (Parvaz et al., 2012) and relative left frontal activity (Papousek et al., 2016). However, another study did not report the effect (Tolegenova et al., 2014). This discrepancy might have been caused by differences in the stimuli and experimental procedure between studies. The present study and the study by Parvaz et al. (2012) both used still images as stimuli, while the study by Tolegenova et al. (2014) presented film clips as stimuli. Given that alpha power is sensitive to the arousal level of stimuli and is decreased by film more than images (for example, see Simons et al., 2003), watching film clips may have dampened the effects of reappraisal on alpha power in the study by Tolegenova et al. (2014). Future studies will be required to verify this possibility.

The results of Experiment 1 also indicated a positive relationship between habitual use of reappraisal and increased relative left frontal activity when instructed to use reappraisal in male participants. The interpretation mentioned above indicates that males who exhibit more habitual use of reappraisal might have shown a greater decrease of emotional responses to negative images when instructed to use reappraisal, compared with those who exhibit less habitual use of

Table 3

Mean frontal alpha asymmetry (FAA) in Experiment 1 and Experiment 2 (standard errors in parentheses). Higher FAA indicates greater relative left frontal activity.

	Experiment 1		Experiment 2	
	Negative	Reappraise Negative	Negative	Suppress Negative
All participants				
Fronto-polar (Fp2 – Fp1)	0.040 (0.013)	0.054 (0.012)	0.034 (0.011)	0.026 (0.011)
Mid-frontal (F4 – F3)	0.094 (0.018)	0.106 (0.016)	0.082 (0.017)	0.084 (0.017)
Lateral-frontal (F8 – F7)	0.002 (0.024)	0.033 (0.024)	–0.018 (0.023)	–0.022 (0.021)
Male participants				
Fronto-polar (Fp2 – Fp1)	0.053 (0.016)	0.078 (0.016)	0.042 (0.014)	0.031 (0.014)
Mid-frontal (F4 – F3)	0.058 (0.024)	0.075 (0.022)	0.026 (0.022)	0.020 (0.023)
Lateral-frontal (F8 – F7)	0.008 (0.030)	0.037 (0.030)	–0.013 (0.028)	–0.022 (0.026)
Female participants				
Fronto-polar (Fp2 – Fp1)	0.027 (0.019)	0.031 (0.019)	0.026 (0.017)	0.021 (0.016)
Mid-frontal (F4 – F3)	0.129 (0.027)	0.137 (0.025)	0.139 (0.025)	0.148 (0.026)
Lateral-frontal (F8 – F7)	–0.003 (0.037)	0.029 (0.036)	–0.023 (0.035)	–0.023 (0.032)

Note. All participants: N = 102; male participants: N = 59; female participants: N = 43.

reappraisal. This supports previous reports that habitual use of reappraisal is positively correlated with decreased emotional responses to negative stimuli reflected in brain activity (Drabant et al., 2009; Moser et al., 2014), as well as one subjective report (Meyer et al., 2012). Habitual use of reappraisal is also related to positive affectivity and better psychological wellbeing (Gross and John, 2003; McRae et al., 2012). Thus, the present finding suggests the possibility that habitual use of reappraisal in everyday life is related to effective emotion regulation. One possible explanation of the finding that habitual use of reappraisal was related to greater relative left frontal activity during instructed reappraisal is that males scoring high on the ERQ reappraisal scale may have been more likely to actually use reappraisal during the laboratory task than low-scorers (just as they are more likely to use reappraisal in everyday life). A previous study reported that individuals differ considerably in the degree to which they adhere to specific emotion regulation instructions, such as using cognitive reappraisal when instructed to do so (Demaree et al., 2006). In addition, previous studies have shown that brain activity during reappraisal is also affected by other individual traits, such as rumination (Ray et al., 2005), mindfulness (Modinos et al., 2010), emotional intelligence (Tolegenova et al., 2014), and reappraisal ability (Papousek et al., 2016). The present study provides additional evidence that habitual use of reappraisal is one of the traits that affects individual differences in neural activity during reappraisal.

Meanwhile, although the results revealed no gender difference in effect of reappraisal instruction on relative left frontal activity, we did not find a significant relationship between habitual use of reappraisal and

increased relative left frontal activity when instructed to use reappraisal in female participants. This finding is in contrast to previous studies that reported a positive relationship between habitual use of reappraisal and decreased emotional responses to negative stimuli in females (Drabant et al., 2009; Moser et al., 2014). A possible explanatory factor is the age difference between participants in the present study and those in previous studies; the participants in the current study were relatively young (19–25 years old) compared with those in previous studies (mean age: 44.0 years old in Drabant et al., 2009; age range: 18–35 years old in Moser et al., 2014). One previous study reported that the use of reappraisal was higher in late middle aged women than in young women (John & Gross, 2004). Thus, the effect of habitual use of reappraisal on brain activity may be more apparent among older females. Future studies are required to examine this hypothesis in detail.

4.2. Effect of suppression on relative left frontal activity

The results of Experiment 2 indicated that relative left frontal activity was no different when participants were instructed to use suppression, compared with normal viewing. Based on the interpretation of the relative left frontal activity mentioned above, this result suggests that being instructed to use suppression might not have decreased emotional responses to negative images. This finding partially supports previous studies reporting that suppression in response to negative stimuli increased responses in emotion-generative regions such as the amygdala and insular (Goldin et al., 2008) and did not change the arousal level

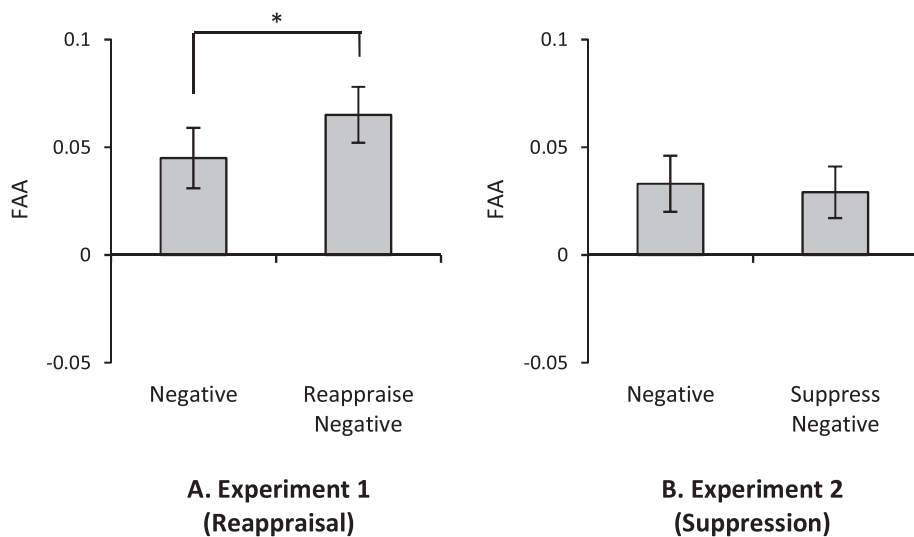


Fig. 1. Comparisons of frontal alpha asymmetry (FAA) between the two conditions in Experiment 1 (A) and Experiment 2 (B) (mean and standard error). Higher FAA indicates greater relative left frontal activity. The FAA was averaged across three pairs, Fp1/2, F3/4, and F7/8. * $p = 0.008$.

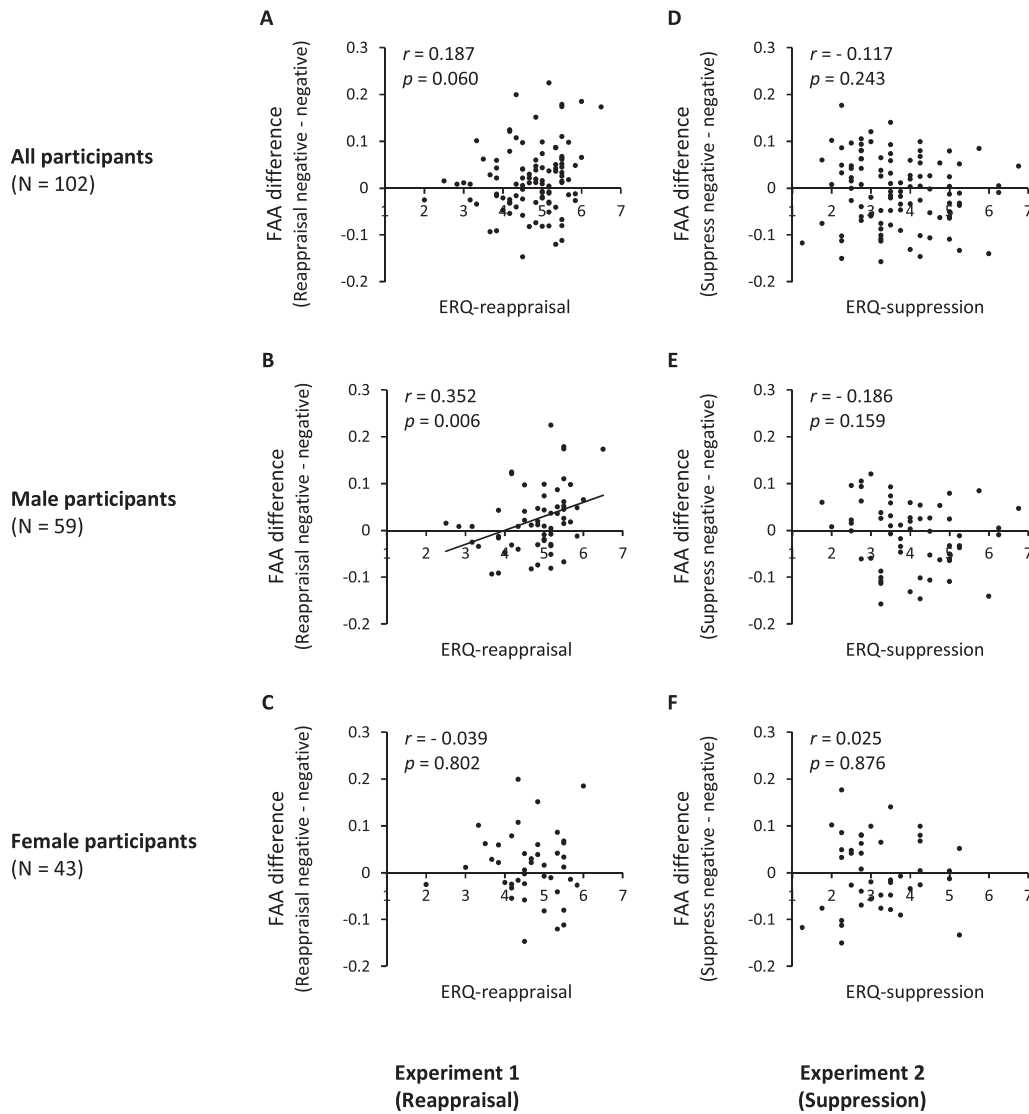


Fig. 2. Correlations between emotion regulation questionnaire (ERQ) scores and frontal alpha asymmetry (FAA) changes in Experiment 1 (A, B, and C) and Experiment 2 (D, E, and F). Left column (A, B, and C) shows the correlation between ERQ-reappraisal scores and the FAA changes by reappraisal (Reappraise Negative minus Negative), while right column (D, E, and F) shows the correlation between ERQ-suppression scores and FAA changes by suppression (Suppress Negative minus Negative). Upper row (A and D) shows the correlation in all participants, middle row (B and E) shows the correlation in male participants, and lower row (C and F) shows the correlation in female participants. Higher FAA indicates greater relative left frontal activity. The FAA was averaged across three pairs, Fp1/2, F3/4, and F7/8.

measured by ERP responses in European Americans (Murata et al., 2013). This is in line with the suggestion that suppression is not an effective emotion regulation strategy (John & Gross, 2004; Augustine and Hemenover, 2009).

It should be noted that cultural differences have been reported in suppression. In the ERP study by Murata et al. (2013) mentioned above, East Asians showed decreased arousal level (measured by late positive potentials) when instructed to use suppression, suggesting that East Asians may be better at regulating emotion through suppression, compared with European Americans. Thus, it is possible that relative left frontal activity was no different between conditions of instructed suppression and normal viewing in the present study because suppression is the normal or habitual way to respond to negative stimuli for people in East Asia. To verify this possibility, further studies are needed to examine relative left frontal activity after instruction to perform suppression among both East Asian and European American participants.

In Experiment 2, we did not observe significant correlations between habitual use of suppression and changes in relative left frontal activity when participants were instructed to use suppression. This

suggests that individuals with high habitual use of suppression did not show different changes in emotional responses to negative images when instructed to use suppression, compared with participants with low habitual use of suppression. This supports the previous finding reported by Meyer et al. (2012) that habitual use of suppression was not related to change of emotional responses to negative stimuli measured by subjective ratings. Thus, the present finding and the previous findings of Meyer et al. (2012) suggest that habitual use of suppression might not affect responses during emotional regulation. However, Walker et al. (2011) found that individuals with high alexithymic traits show a greater ability to suppress emotional responses in response to negative stimuli than those with low alexithymic traits. Given alexithymia is positively related to habitual use of suppression (for example, Swart et al., 2009), further studies are needed to examine the relationships between habitual use of suppression, emotional responses during suppression, and other personality traits such as alexithymia. In addition, habitual use of suppression is related to poorer psychological wellbeing (Gross and John, 2003; McRae et al., 2012). Thus, further studies are needed to examine the effects of habitual use of suppression on long-term emotional responses.

4.3. Limitations

The present study contained several limitations. First, we did not measure behavioral responses (for example, vocalizing the idea of reappraisal, Papousek et al., 2016; writing the idea of reappraisal, Wang et al., 2015). Thus, we could not verify whether participants actually reappraised and suppressed negative emotion or not. As such, it is unclear whether participants were distracted from the task, although we instructed them to avoid thinking about unrelated things while watching images. Future studies are needed to examine relative left frontal activity during reappraisal and suppression, adopting an experimental procedure that addresses this limitation.

Second, we did not measure subjective ratings (for example, emotional intensity and effort expenditure, Moser et al., 2014) and thus could not examine whether being instructed to use reappraisal and suppression affected subjective responses, or whether subjective responses are related to relative left frontal activity. Although physiological responses provide more direct indices of responses (for example, Schwerdtfeger, 2004), future studies are needed to examine relative left frontal activity during reappraisal and suppression by measuring both psychological and physiological responses.

Finally, the reliability of ERQ-reappraisal in the present study was low (Cronbach's alpha: 0.686), and thus this might have affected the results regarding the relationship between habitual use of reappraisal and relative left frontal activity. In future studies, in addition to the ERQ, other measures are required as indices of personality traits related to emotion regulation.

5. Conclusion

The present results of relative left frontal activity suggest that emotional responses to negative stimuli are decreased when participants are instructed to use reappraisal, but not when instructed to use suppression. Moreover, our findings suggest the possibility of a positive relationship between habitual use of reappraisal and decreased emotional responses to negative stimuli when instructed to use reappraisal in males. Taken together, these findings support previous reports that reappraisal is a more effective emotion regulation strategy than suppression.

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Appendix A. International Affective Picture System (IAPS) picture identification numbers

Experiment 1

Neutral images: 2190, 2200, 2210, 2230, 2570, 2840, 5500, 5531, 5950, 7000, 7002, 7009, 7010, 7020, 7025, 7035, 7050, 7080, 7100, 7150, 7160, 7170, 7175, 7190, 7217, 7224, 7233, 7235, 7550, 7700.

Negative images: 2683, 2688, 2811, 3170, 3181, 3220, 3301, 3350, 3500, 3530, 6212, 6230, 6231, 6242, 6312, 6313, 6520, 6550, 6821, 6825, 8485, 9050, 9183, 9414, 9425, 9428, 9620, 9622, 9903, 9908.

Experiment 2

Neutral images: 2214, 2221, 2273, 2383, 2512, 5471, 6150, 7001, 7003, 7012, 7016, 7017, 7019, 7021, 7037, 7041, 7055, 7057, 7058, 7059, 7090, 7161, 7165, 7185, 7211, 7255, 7500, 7705, 7950, 9260.

Negative images: 1300, 2053, 2710, 2800, 3080, 3160, 3230, 6200, 6210, 6244, 6250, 6260, 6370, 6540, 6560, 6570, 6571, 9040, 9163, 9250, 9410, 9412, 9413, 9490, 9520, 9600, 9635.1, 9911, 9920, 9921.

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